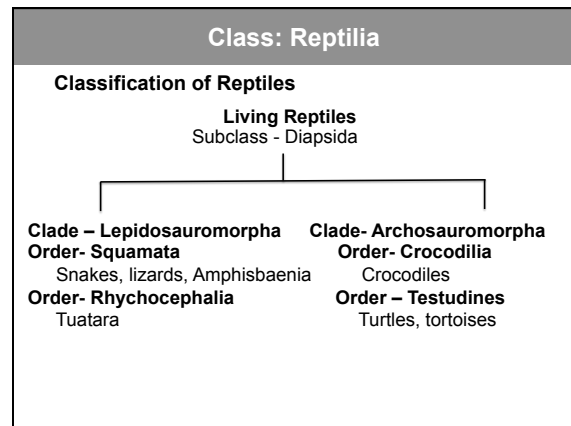
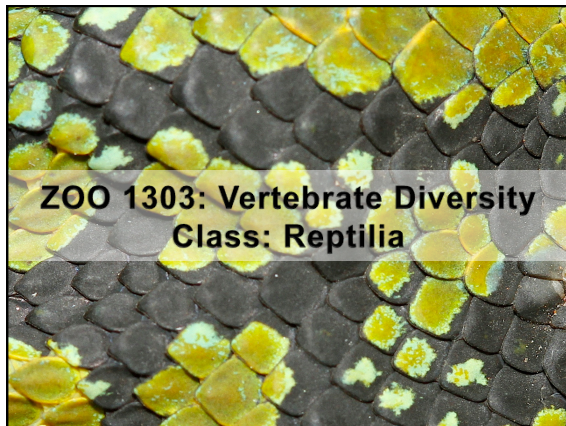




Class: Reptilia

Order- Crocodilia: Crocodiles & Alligators

- Arose in the Cretaceous period (145 - 65 Mya)
- Have remained mostly unchanged for nearly 150 My
- Have an elongate, robust, well-reinforced skull
- Massive jaw musculature - provide a wide gape and rapid, powerful closure
- Teeth are set in sockets – **Thecodont**

Class: Reptilia

Order- Crocodilia: Crocodiles & Alligators

- Complete secondary palate present in the mouth– allows to breathe when the mouth is filled with water or food
- Four chambered heart
- Size – 1.5 m to 6 m, weighing up to 1000 kg
- Scutes in the skin are supported by bony plates - osteoderms

Class: Reptilia

Order- Crocodilia: Crocodiles & Alligators

- Oviparous - Usually 20 to 50 eggs are laid in a mass of dead vegetation and guarded by the mother
- Temperature dependent sex determination
- Found in fresh and brackish water habitats in the tropical and subtropical regions in the world

Class: Reptilia

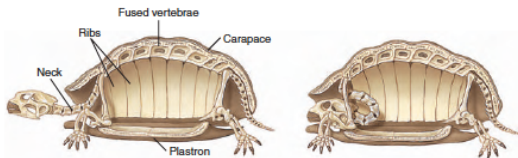
Order- Crocodilia: Crocodiles & Alligators

- Includes Alligators, Crocodiles, Caimans, Gharial

Class: Reptilia

Order- Testudines: Turtles & Tortoises

- Appear in the fossil record from the late Triassic, ~200 Mya
- Very little change in morphology since then
- Bodies are enclosed in shells consisting of a dorsal **carapace** and a ventral **plastron**
- Shell composed of two layers: outer horny layer of keratin
inner layer of bone
- Bony layer- fusion of ribs, vertebrae



Class: Reptilia

Order- Testudines: Turtles & Tortoises

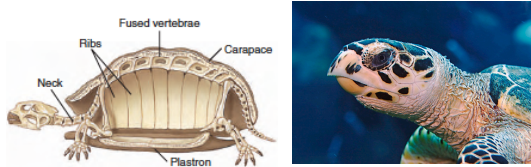
- Includes the modern day turtles, tortoises, terrapins etc.
- Size: 8cm – 2m



Class: Reptilia

Order- Testudines: Turtles & Tortoises

- Limbs and limb girdles are located inside the rib cage
- Lacks teeth, jaw is provided with tough, horny plates for gripping food



Class: Reptilia

Order- Testudines: Turtles & Tortoises

- Rigid shell with fused ribs - cannot expand its chest to breathe
- Air is taken in by increasing abdominal cavity volume by contracting posterior abdominal muscles (active process)
- Exhalation accomplished by contracting the pectoral girdle muscles compressing the viscera and forcing air out of the lungs (also an active process)

Class: Reptilia

Order- Testudines: Turtles & Tortoises

- Brain- small, does not exceed 1% of the body weight
- Both middle and an inner ear present, perception of sound is poor
- Have a good sense of smell, vision and color perception
- Slow moving
- Diet – herbivores, omnivores and carnivores

Class: Reptilia

Order- Testudines: Turtles & Tortoises

- Reproduction – oviparous, internal fertilization, bury their shelled eggs in the ground
- Sex – determined by the temperature of the nest
Low temperatures during incubation - produce males
High temperatures during incubation - produce females



Class: Reptilia

Order- Rhychocephalia: Tuataras

- Has Triassic origins (252-201 Mya)
- Has numerous features that are identical to 200 My old fossils
- Primitive diapsid skull – complete temporal bar



Tuatara



Modern diapsid



Class: Reptilia

Order- Rhychocephalia: Tuataras

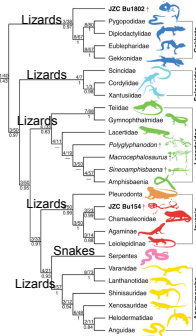
- Hemipenis not fully developed
- Lizard like body form, length 66 cm or less
- Represented by the single species *Sphenodon punctatus*
- Restricted to small islets in New Zealand
- Lives in burrows



Class: Reptilia

Order- Squamata: Lizards, Snakes & Worm Lizards

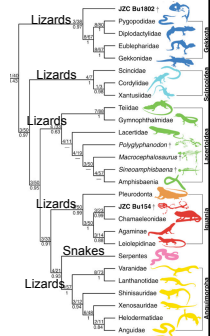
- Most recent and diverse of the diapsid evolution
- Includes ~95% of all known living reptiles
- Lizards appeared in the Permian period (300 -250 MYA)
- Began their radiation in Cretaceous period (145-66 MYA)



Class: Reptilia

Order- Squamata: Lizards, Snakes & Worm Lizards

- Worm lizards, have structural specializations associated with burrowing
- Snakes evolved from lizards during the late Cretaceous period
- Snakes evolved through extreme elongation of the body and displacement and rearrangement of internal organs



Class: Reptilia

Order- Squamata: Lizards, Snakes & Worm Lizards

	Lizards	Snakes
Limbs	Present/ Absent	Absent
Body Shape	Maybe elongated	Elongated
Eye-lids	movable or fixed	No eye-lids
Scales	overlapping or non-overlapping	overlapping
External ear openings	present or absent	Absent
Pelvic/pectoral girdle	Present	Present /Absent
Lungs	Two	One functional

Class: Reptilia

Order- Squamata: Lizards, Snakes & Worm Lizards

Reproduction

- Shows both oviparity and ovoviviparity
- Oviparity: laying of fertilized eggs
- Ovoviviparity: retention of eggs in side the mother and giving birth to live young



Class: Reptilia

Order- Squamata: Lizards, Snakes & Worm Lizards

Reproduction

- Ovoviviparity in reptiles is seen only among squamates
- Ovoviviparity evolved at least 100 separate times - associated with cold climates
- Ovoviviparity was achieved through increasing the length of time the eggs are kept within the oviduct
- Developing young respire through extraembryonic membranes and obtain nutrition from - yolk sacs (**lecithotrophy**)

Class: Reptilia

Order- Squamata: Worm Lizards

- Have elongate, cylindrical bodies
- Most lack external limbs
- Highly specialized burrowing forms
- Skin is soft and is divided into numerous rings
- Eyes and ears are hidden under skin - look like earthworms
- Distribution - South America and Southern North America and tropical Africa

Class: Reptilia

Order- Squamata: Worm Lizards



Class: Reptilia

Order- Squamata: Typical Lizards

- Extremely diverse group
- Terrestrial, burrowing, aquatic, and arboreal members
- Majority of lizards have four limbs and relatively short bodies
- In some, the limbs are degenerated or completely absent
- Movable eyelids
- External ear present
- Live in warm tropical and subtropical regions of the world
- Body temperature is kept relatively constant by behavioral thermoregulation

Class: Reptilia

Order- Squamata: Typical Lizards

Includes: Iguanas, monitor lizards, Chameleons, dragon lizards, geckoes, skinks



Dragon lizard



Gecko



Skink (limbless)

Class: Reptilia

Order- Squamata: Typical Lizards

Includes: Iguanas, monitor lizards, Chameleons, dragon lizards, geckoes, skinks



Iguanas



Chameleon



Monitor lizards

Class: Reptilia

Order- Squamata: Snakes

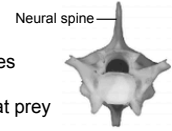
- Entirely limbless and lack both pectoral and pelvic girdles (exception pythons, boas)
- Vertebrae numerous shorter and wider than those of tetrapod reptiles – allows quick lateral undulations
- Ribs increase rigidity of the vertebral column



Class: Reptilia

Order- Squamata: Snakes

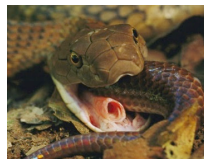
- Neural spine elevated - gives the muscles more leverage
- Highly **kinetic skull** – enable them to eat prey several times their own diameter
- Two halves of the lower jaw (mandibles) are joined only by muscles and skin
- Many of the skull bones are so loosely articulated (i.e. fitted)



Class: Reptilia

Order- Squamata: Snakes

- Tracheal opening (glottis) is found between the two mandibles
- Cornea of the eye is permanently protected with a transparent membrane – spectacle
- Fixed eyelids
- Vision is poor, arboreal snakes are an exception
- No external ears or tympanic membrane



Class: Reptilia

Order- Squamata: Snakes

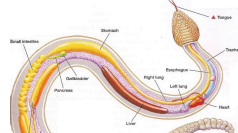
- Have internal ears (stapes present), hear low frequencies (100 to 700 Hz)
- Sensitive to vibrations carried in the ground
- Heavily depend on chemical senses - **Jacobson's organs**
- Some have heat-sensitive organs positioned in pits on their heads – pit organs (e.g. pit-vipers, pythons)



Class: Reptilia

Order- Squamata: Snakes

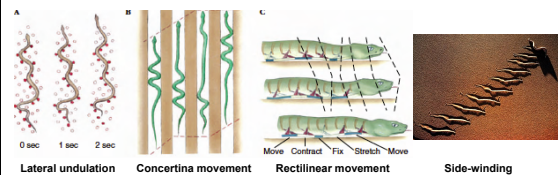
- Body is entirely covered with a tough, impervious skin
- Skin is covered with overlapping scales
- Extreme elongation - displacement and rearrangement of internal organs (one lung or left lung is reduced)



Class: Reptilia

Order- Squamata: Snakes

- Movement
 - Lateral undulation** - S-shaped path
 - Concertina movement** - S-shaped loops
 - Rectilinear movement** - straight line
 - Side-winding** – throwing body forward in loops



Class: Reptilia

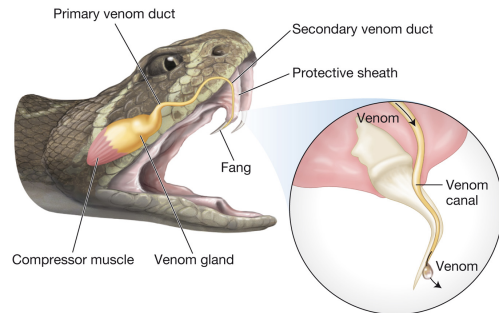
Order- Squamata: Snakes

- Some have a pair of teeth on the maxillary bones modified as fangs
- Fangs deliver venom that helps in immobilizing and digestion of prey
- Venom is stored in venom glands in the head, modified salivary glands
- Venom mixture of different types of,
 - Enzymes (90-95%)
 - neurotoxins (inhibits nerve transmission)
 - hydrolytic enzymes (digest cell membranes)
 - Polypeptides - Cytotoxins
 - Inorganic ions



Class: Reptilia

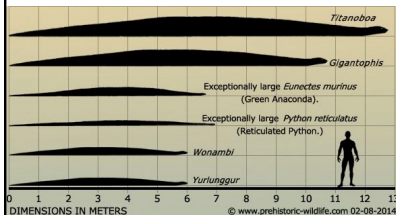
Order- Squamata: Snakes



Class: Reptilia

Order- Squamata: Snakes

Size – 10cm – 7m



Smallest snake in the world- Barbados thread snake (10 cm)



- Large extinct snakes – *Titanoboa*, *Gigantophis*, *Wonambi*, *Yurlunggur*

Class: Reptilia

Order- Squamata: Snakes



Class: Reptilia

Order- Squamata: Snakes

